Cellular organization of inflammatory transcription in the mouse cochlea after acoustic trauma

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Abstract

Acoustic trauma is associated with divergent inflammatory transcription across cochlear stromal, vascular and immune populations. We analyzed replicated single-cell profiles alongside whole-cochlea transcriptomes and complementary expression-localization data from public mouse datasets. Fibrocyte/mesenchymal populations showed tumor necrosis factor–nuclear factor κB (TNF–NF-κB) enrichment toward noise exposure in the lateral wall and spiral ganglion region. Endothelial and immune/myeloid populations showed enrichment toward the unexposed condition. These enrichment directions persisted after omission of individual animals from the models. Chemokine-associated transcription was prominent in the lateral-wall fibrocyte response. Lateral-wall endothelial cells showed mitogen-activated protein kinase (MAPK) family enrichment with a prominent proteasome component. Whole-cochlea profiles showed inflammatory, protein-folding and unfolded-protein-response enrichment after exposure. A subset of candidate genes retained recurrent endothelial or fibrocyte/mesenchymal mean-expression maxima across pooled condition libraries, including under the alternative cluster annotation. Stromal inflammatory and chemokine transcription coexists with vascular protein-quality-control programs in the cochlear response to noise.

Keywords: acoustic trauma; cochlea; single-cell RNA sequencing; pseudobulk; fibrocytes; NF-κB

Introduction

The cochlea converts acoustic vibration into neural signals through coordinated sensory, neuronal, vascular and supporting-cell functions. Acoustic overexposure perturbs this multicellular system and initiates transcriptional responses that extend beyond sensory hair cells. Experimental noise exposure can leave persistent cochlear synaptic and neural injury after hearing thresholds recover, with differential vulnerability among auditory nerve fibers [1, 2]. Acoustic injury also induces cochlear cytokine expression and recruits mononuclear phagocytes, linking tissue stress with local immune responses [3, 4].

Single-cell atlases resolve the molecular diversity of cochlear supporting, neural and lateral-wall populations across postnatal maturation and aging [5, 6]. Spiral ganglion neurons comprise molecularly distinct subclasses with different afferent connectivity [7]. Single-cell profiling has established distinct neuronal and lateral-wall responses to acoustic trauma, including stress-response programs and changes in ion-transport genes [8]. Whole-cochlea proteomic and transcriptomic analyses have also connected noise exposure with proteotoxic stress and recruitment of the proteostasis network [9].

A central question is how a tissue-level inflammatory signature is distributed among the cells that contribute to it. A pathway can be preferentially expressed in one cell population while changing in different directions across populations after exposure. Earlier reporter studies identified noise-associated nuclear factor κB (NF-κB) activity in cochlear fibrocytes and linked the response to cell type and mouse strain [10, 11]. Replicated transcriptomic datasets now permit sample-level estimates of these cellular expression responses. Such estimates retain the biological sample as the unit of inference, following established frameworks for replicated single-cell differential-state analysis [12, 13].

We asked whether cochlear cell populations contributing to inflammatory gene expression showed similar or opposing responses after acoustic trauma. GSE168041 provided replicated spiral ganglion and lateral-wall profiles for estimating cell-type-specific responses [8]. GSE160639 provided whole-cochlea measurements across three sound levels [9], and GSE252285 provided expression localization across control, hidden hearing loss and noise-induced hearing loss conditions [14]. Animal-level pathway estimates and leave-one-animal-out analyses resolve cell-type inflammatory divergence, complemented by annotation sensitivity of the expression-localization patterns.

Results

The replicated GSE168041 profiles comprised 64,705 cells from 16 samples, including 31,556 spiral ganglion and 33,149 lateral-wall cells (Fig. 1; Supplementary Fig. S1). Thirteen tissue–cell-type combinations retained four unexposed and four noise-exposed samples. Fibrocyte/mesenchymal, endothelial and immune/myeloid populations were represented in both compartments. Sample-level retention after quality control and doublet removal ranged from 86.6% to 92.6%; differential-expression counts for all thirteen profiles are shown in Supplementary Fig. S2.

NF-κB responses diverge across cochlear cell types

TNF–NF-κB enrichment differed in direction across cochlear cell types (Fig. 2). Fibrocyte/mesenchymal cells showed enrichment toward noise exposure in the lateral wall (NES 1.648; FDR 0.00371) and spiral ganglion region (NES 1.463; FDR 0.0305). In the lateral wall, the 70-gene leading edge included the chemokine gene *Ccl5*, which had a $\log_2$ fold change of 5.958 and FDR

4.12×10^{-14}. Other leading-edge members included *Ifih1*, *Mcl1* and *Ifit2*, with $\log_2$ fold changes of 1.321, 0.467 and 2.061, respectively, and FDR values of 0.00227, 0.0209 and 0.0375.

Endothelial and immune/myeloid populations showed enrichment of the same gene set toward the unexposed condition in both regions. In the lateral wall, endothelial cells had NES −2.813 and FDR 4.27×10^{-20}, and immune/myeloid cells had NES −3.213 and FDR 1.78×10^{-32}. In the spiral ganglion preparation, the corresponding NES values were −1.854 and −2.375, with FDR values of 7.62×10^{-5} and 1.38×10^{-12}. Positive TNF–NF-κB enrichment also occurred in spiral ganglion neurons (NES 1.820; FDR 5.17×10^{-5}) and lateral-wall epithelial/supporting-associated cells (NES 2.143; FDR 1.31×10^{-8}).

Lateral-wall IL6–JAK–STAT3 signaling was enriched toward the unexposed condition in endothelial cells (NES −1.796; FDR 0.00331) and immune/myeloid cells (NES −1.952; FDR 0.000362). Endothelial MAPK family signaling cascades were enriched toward noise exposure (NES 1.918; FDR 8.37×10^{-7}). Its 48-gene leading edge contained 28 proteasome-subunit genes, together with *Map2k2*, *Mapk3* and *Dusp2*. Endothelial *Map2k2* and *Mapk3* had $\log_2$ fold changes of 0.417 and 0.336, with FDR values of 0.000427 and 0.00466. Endothelial angiogenesis and VEGF enrichment did not meet the FDR criterion, with NES values of 1.173 and −1.061 and FDR values of 0.384 and 0.513.

Opposing enrichment directions persisted after individual-animal omission

All 56 focused pathway estimates retained the direction of the full-sample analysis across the 48 leave-one-animal-out fits (Fig. 3). Fibrocyte/mesenchymal TNF–NF-κB enrichment remained positive in all eight omissions in each region. Spiral-ganglion estimates ranged from NES 1.305 to 1.731, with FDR < 0.05 in seven of eight fits. Lateral-wall estimates ranged from NES 1.317 to 2.007, with six of eight fits meeting this criterion. FDR ranges were 0.000326–0.136 and 3.94×10^{-6}–0.169, respectively. Endothelial and immune/myeloid TNF–NF-κB enrichment remained negative and significant in all eight omissions in each region.

Lateral-wall endothelial MAPK-family enrichment remained positive and significant in all eight omissions, with NES 1.807–1.925 and FDR 5.50×10^{-7}–2.08×10^{-5}. Individual-animal pathway profiles are shown alongside the omission estimates in Fig. 3; expression profiles for the reported leading-edge genes are shown in Supplementary Fig. S3.

Whole cochleae showed inflammatory and proteostatic enrichment at both exposure levels

Whole-cochlea profiles comprised 19,673 genes from twelve samples, with four samples at each of 70, 94 and 105 dB (Fig. 4). At FDR < 0.05, 5,777 genes differed between 94 and 70 dB, 889 between 105 and 70 dB, and 4,177 between 105 and 94 dB (Supplementary Fig. S4). The overall dose likelihood-ratio test identified 6,398 genes.

TNF–NF-κB, protein-folding and unfolded-protein-response gene sets were positively enriched at both 94 and 105 dB relative to 70 dB (Fig. 4). Overall, 17 of 27 pathway–contrast combinations met global FDR < 0.05: eight for 94 versus 70 dB and nine for 105 versus 70 dB. For TNF–NF-κB signaling, the corresponding NES values were 2.563 and 2.979, with global FDR values of 3.45×10^{-18} and 4.41×10^{-24}. At 94 dB, the unfolded protein response had NES 2.059 and global FDR 3.75×10^{-6}. At 105 dB, this pathway had NES 2.512 and global FDR 5.17×10^{-10}. Protein folding yielded NES 1.800 and 2.155, respectively, with global FDR values of 2.80×10^{-5} and 3.06×10^{-8}.

None of the nine targeted pathways met the global FDR criterion in the direct 105-versus-94-dB comparison. TNF–NF-κB signaling yielded NES -1.135 and global FDR 0.223, while the reactive oxygen species pathway yielded NES -1.488 and FDR 0.0595. The two exposed groups shared positive inflammatory enrichment relative to the 70-dB reference, while retaining distinct genome-wide profiles. Positive whole-cochlea TNF–NF-κB enrichment accompanied the fibrocyte-associated response observed in the separately sampled GSE168041 dataset.

Eleven selected genes retained endothelial or fibrocyte localization across three libraries

Eleven of the twenty selected genes had the same highest-expression broad cell category across all three GSE252285 libraries (Fig. 5; Supplementary Fig. S5). Categories were ranked by mean normalized expression, and each corresponding group contained at least 20 cells. The 36,111 cells comprised 10,273 control, 16,631 hidden hearing loss and 9,207 noise-induced hearing loss profiles.

The seven genes with recurrent endothelial expression maxima were *Dusp1*, *Nfkbia*, *Apold1*, *Sik1*, *Fosb*, *Rasd1* and *Depp1*. The four genes with recurrent fibrocyte/mesenchymal expression maxima were *Ccn1*, *Errfi1*, *Nr4a1* and *Egr1*. Endothelial *Nfkbia* detection fractions were 87.5%, 85.3% and 87.3% in control, hidden hearing loss and noise-induced hearing loss libraries, respectively. Fibrocyte/mesenchymal *Egr1* detection fractions were 89.5%, 91.5% and 87.3%.

The eleven recurrent cell-type localizations persisted when cluster 19 was included in the fibrocyte/mesenchymal group (Fig. 6; Supplementary Fig. S6). *Cebpb* additionally acquired a recurrent fibrocyte/mesenchymal maximum. Cluster 19 expressed fibrocyte-associated markers, including *Coch* in 99.7% and *Otor* in 82.1% of its 319 cells. *Pecam1* and *Kdr* were detected in 1.9% and 9.1%, respectively, and *Vwf* was not detected.

Apold1 was highly expressed in both endothelial cells and cluster 19 (Supplementary Fig. S7). Its mean normalized expression was 1.585 in cluster 19 and 1.338 across endothelial cells, with detection fractions of 75.9% and 66.9%, respectively. Endothelial cells retained the highest broad-cell-category mean in all three libraries after cluster 19 was included in the fibrocyte/mesenchymal group.

Five of the six UCell-guided follow-up comparisons exceeded the expression- and detection-matched null at the six-test FDR threshold (Supplementary Fig. S8). The endothelial and fibrocyte/mesenchymal TNF–NF-κB localization statistics exceeded the matched null distributions, with observed score differences of 0.0545 and 0.0380 and FDR 0.000200 for both tests. Endothelial VEGF/angiogenesis and endothelial and immune/myeloid IL6–JAK–STAT3 localization also met the six-test FDR criterion. The endothelial MAPK/AP-1 localization statistic yielded a score difference of 0.0176 and FDR 0.0724. Adding gene-length strata retained the five primary-positive localization results. Reassigning cluster 19 preserved all six significance classifications in both matching analyses (Supplementary Fig. S6). The fibrocyte/mesenchymal TNF–NF-κB score difference increased from 0.0380 to 0.0394, with FDR 0.000200 in both annotations. Endothelial MAPK/AP-1 remained non-significant, with primary FDR 0.0725 and gene-length-matched FDR 0.0817 under the alternative annotation.

In GSE168041, lateral-wall endothelial *Apold1* had a $\log_2$ fold change of 0.031 and FDR 0.913. Fibrocyte/mesenchymal *Errfi1* had a $\log_2$ fold change of -0.361 and FDR 0.588.

Discussion

Whole-cochlea inflammatory enrichment coexists with opposing responses in stromal, vascular and immune populations. TNF–NF-κB enrichment favored noise exposure in fibrocyte/mesenchymal populations from both dissected regions. Endothelial and immune/myeloid populations showed the opposite direction, with neuronal and epithelial/supporting-associated populations adding regional variation. Animal omissions preserved these directions; fibrocyte FDR significance varied, while endothelial and myeloid enrichment remained significant.

Milon and colleagues established cell-type-specific transcriptional responses to acoustic trauma [8]. The fibrocyte response complements reporter evidence of noise-associated NF-κB activity in spiral-ligament fibrocytes [10, 11]. The lateral-wall *Ccl5* increase adds a chemokine-associated component to this response. Previous experiments detected cytokine production in spiral-ligament fibrocytes and chemokine and adhesion-molecule expression during leukocyte recruitment [3, 15]. Together, these observations connect stromal inflammatory transcription with genes associated with local immune communication.

Myeloid TNF–NF-κB enrichment was negative at 24 hours, within a temporally and anatomically heterogeneous immune response. Acoustic injury recruits cochlear mononuclear phagocytes and induces antigen-presentation features in basilar-membrane phagocytes [4, 16]. Resident macrophage responses have been linked to hair-cell injury and TLR4 signaling [17]. Fate mapping distinguishes the spatial and temporal distributions of resident and recruited macrophages after trauma [18]. Selected innate-immune genes increased in specific myeloid populations in the original study [8]. Their induction can coexist with negative enrichment of the broader TNF–NF-κB set. Cell-state changes and subtype composition may both contribute to this program-specific myeloid response.

Proteasome-subunit genes comprised 28 of 48 MAPK-family leading-edge members in lateral-wall endothelial cells, alongside *Map2k2*, *Mapk3* and *Dusp2*. This composition links the annotated signaling program to a substantial protein-quality-control transcriptional response. Whole-cochlea protein-folding and unfolded-protein-response enrichment fits the proteotoxic response described by Jongkamonwiwat and colleagues [9]. Independent experiments have measured time-dependent MAPK phosphorylation [19] and delayed cochlear oxidative damage after acoustic trauma [20]. NRF2 loss and pharmacological activation also link antioxidant defense to acoustic vulnerability in mice [21].

Endothelial stress responses also connect to the cellular architecture of the cochlear barrier. Interactions among cochlear endothelial cells, pericytes and perivascular macrophage-like melanocytes regulate strial permeability [22]. Noise exposure can alter perivascular contacts and endothelial junctions in this compartment [23].

Expression localization adds cell-type context to condition-associated changes across cochlear populations. Endothelial and fibrocyte/mesenchymal TNF–NF-κB localization in GSE252285 co-existed with opposite enrichment directions in GSE168041. *Apold1* expression spanned endothelial cells and fibrocyte-associated cluster 19, retaining an endothelial maximum in mean expression across broad cell types. Reassigning cluster 19 to fibrocytes preserved the eleven recurrent localization patterns and all matched-null significance classifications. *Apold1* and *Errfi1* thus illustrate recurrent localization despite nonsignificant condition effects in the replicated profiles.

The datasets sample distinct exposure windows and anatomical settings in the cochlea. GSE160639 sampled whole cochleae immediately after a 30-minute exposure, whereas GSE168041 profiled dissected compartments 24 hours after a two-hour exposure [8, 9]. GSE252285 offers descriptive localization from one pooled library per condition, collected six hours after the final exposure [14]. Its UCell-guided comparisons remain exploratory because the six-test FDR family

excludes the preceding program–cell-type screen. Documented strain-dependent fibrocyte NF-κB responses also inform comparisons across these experimental settings [11].

Fibrocyte-associated TNF–NF-κB and chemokine transcription form a stromal component of the cochlear response to acoustic trauma. Endothelial and immune/myeloid populations show opposing enrichment of the same inflammatory gene set. Proteasome-rich endothelial MAPK enrichment and whole-cochlea protein-folding responses place protein quality control alongside inflammation in the cochlear stress response.

Methods

Ethics statement

This study analyzed publicly available mouse sequencing data and involved no new animal procedures. Animal procedures are described in the source publications and dataset records [8, 9, 14]. Approvals for the published experiments are reported in the original articles.

Public datasets and study design

Three mouse datasets were examined. GSE168041 contains spiral ganglion and lateral-wall single-cell RNA-sequencing profiles from unexposed and noise-exposed male CBA/CaJ mice [8]. Noise exposure comprised 105 dB sound pressure level for two hours, with tissue collection 24 hours later. Spiral ganglion preparations were obtained from nine-week-old mice and lateral-wall preparations from seven- to eight-week-old mice. Each compartment contributed four mice per condition.

Accession-specific metadata for GSE160639 specify postnatal-day-60 FVB/NCrl mice exposed to 70, 94 or 105 dB for 30 minutes, with four whole-cochlea samples per group and immediate collection [24]. The source publication reports an equal overall representation of male and female mice [9]; sex was not encoded in the deposited sample-level metadata.

GSE252285 contains three single-cell libraries from C57BL/6J mice corresponding to control, hidden hearing loss and noise-induced hearing loss [14]. The deposited sample titles specify six mice for each library. Hidden hearing loss exposure used 110 dB for two hours; noise-induced hearing loss exposure used 115 dB for four hours on each of three days. Collection occurred six hours after the final exposure. Sex was not provided in the sample records.

Selection of genes and pathway comparisons

The localization analysis used an author-selected exploratory panel of twenty genes, with membership retained across all three GSE252285 libraries. The bulk analysis evaluated nine targeted gene sets covering cytokine and inflammatory signaling, proteostasis, oxidative stress and apoptosis.

Initial UCell profiles of six programs in the same cells guided six pathway–cell-type comparisons for exploratory matched-gene-set analysis. Selection sources and exact definitions are summarized in Supplementary Table S1.

Quality control and cell annotation in GSE168041

The 16 deposited count matrices contained 71,743 cells and 27,998 Ensembl gene identifiers. Cell barcodes were prefixed with their sample accession, and raw nonnegative integer counts were retained separately from the normalized analysis matrix. Cells were retained with 500–7,500 detected

genes, a mitochondrial fraction of at most 30% and a hemoglobin fraction of at most 20%. Quality filtering retained 69,036 cells before doublet detection. Doublets were identified separately for each sample using scDblFinder 1.23.4 [25]. The random seed was 20260825 plus the sample index.

Spiral ganglion and lateral-wall preparations were processed separately using Scanpy 1.12.1 and AnnData 0.12.16 [26]. Counts were normalized to 10,000 per cell and log-transformed. Highly variable gene selection used 3,000 genes with the Seurat flavor and sample accession as the batch key. Scaled values were capped at 10. Principal component analysis retained 50 components, and the neighbor graph used 15 neighbors and 40 principal components. UMAP used a minimum distance of 0.5, and Leiden clustering used resolution 0.6 [27]. The random seed was 20260825.

Broad cell annotations were assigned from canonical cochlear markers and cluster-marker profiles. Candidate genes and the inflammatory programs examined subsequently were excluded as deciding annotation evidence. Annotations were fixed before differential-expression analysis. Rare or conflicting clusters were retained in composition summaries as unassigned. Computational batch correction was not applied.

Sample level pseudobulk differential expression

Raw counts were summed within each sample and broad cell category. Eligibility required at least three samples per condition with at least 20 cells each, positive library sizes, a resolved annotation and a maximum sample contribution of 50% of the condition-specific cells. Thirteen tissue–cell-type combinations met these criteria, each with four samples per condition.

Differential expression was estimated separately for each eligible combination using PyDESeq2 0.5.4 with the design formula $\sim$ condition [28, 29]. The contrast was noise-exposed versus unexposed, so positive $\log_2$ fold changes indicated higher expression after exposure. Genes with fewer than 10 total counts were removed before model fitting. Cook's-distance filtering and independent filtering were enabled, the target FDR was 0.05, and $\log_2$ fold-change shrinkage was not applied. The biological sample was the unit of inference [12, 13].

Gene set enrichment in cell type profiles

Genes with finite Wald statistics and available symbols were ranked by descending Wald statistic for preranked gene-set enrichment [30]. Duplicate symbols were resolved by maximum absolute statistic, followed by maximum signed statistic and lexical Ensembl identifier.

Mouse MSigDB 2026.1.Mm gene sets were obtained using msigdbr 26.1.1 [31]. The analysis combined the Hallmark collection [32] and Reactome pathway annotations [33]. Enrichment used fgseaMultilevel in fgsea 1.36.2, a weighting exponent of 1, gene-set sizes of 10–500, eps = 0 and seed 20260825.

Benjamini–Hochberg adjustment was applied across the combined eligible Hallmark and Reactome sets within each cell-type model [34]. Positive and negative NES values indicated enrichment toward noise-exposed and unexposed samples, respectively.

The focused panel comprised TNF signaling via NF-κB, IL6–JAK–STAT3 signaling, angiogenesis, VEGF signaling, MAPK family signaling cascades and activation of the AP-1 family of transcription factors. Leading-edge membership identified the genes contributing to the enrichment-score extremum.

Sensitivity to individual biological samples

Leave-one-animal-out analyses evaluated fibrocyte/mesenchymal, endothelial and immune/myeloid populations in each dissected compartment. Each of the six models was refitted eight times, retaining three samples from one condition and four from the other. Each fit retained the original model's gene universe and re-estimated normalization and dispersion with the same PyDESeq2 settings.

Enrichment retained the original eligible Hallmark–Reactome testing family for each tissue–cell-type model. Any unavailable pathway remained in the Benjamini–Hochberg family denominator. Ranked inputs used gene symbol and gene identifier as secondary ordering keys for tied Wald statistics. TNF–NF-κB estimates were summarized across all six models, together with MAPK-family estimates in lateral-wall endothelial cells. Direction, NES and FDR were summarized for every omission. Omission ranges summarize overlapping seven-animal subsets and quantify sample influence within the original cohort.

Individual-animal profiles used normalized pseudobulk counts from the full eight-sample models. Pathway profiles averaged gene-wise z scores of log1p-normalized counts across the fixed pathway members. Standardization used the mean and sample standard deviation across all eight animals, with zero-variance genes omitted. Duplicate symbols followed the full-model ranking rule. These descriptive profiles were displayed alongside normalized expression of the reported leading-edge genes.

Whole cochlea differential expression and pathway summaries

The public GSE160639 read-count matrix contained 35,496 genes across 12 samples. The analysis included 19,673 genes and all twelve samples, with four samples per dose. The gene universe and Ensembl-to-symbol mapping were retained from the earlier bulk analysis.

R 4.6.1 and DESeq2 1.52.0 estimated median-ratio size factors within this gene universe [28]. A categorical dose model used 70 dB as the reference and a parametric dispersion trend. Wald tests evaluated 94 versus 70 dB, 105 versus 70 dB and 105 versus 94 dB. The likelihood-ratio test compared the dose model with an intercept-only model.

Independent filtering and Cook's-distance filtering were enabled, with target FDR 0.05 and Benjamini–Hochberg adjustment within each gene-level testing family. Neither a coefficient prior nor fold-change shrinkage was applied. Outlier replacement required at least seven replicates per group; each dose group contained four. Fitting used one process and seed 20261101.

Targeted enrichment used the signed Wald statistics from the three bulk contrasts. Four GO Biological Process sets represented cytokine-mediated signaling, protein folding, response to endoplasmic reticulum stress and response to oxidative stress. Five Hallmark sets represented apoptosis, inflammatory response, reactive oxygen species pathway, TNF signaling via NF-κB and unfolded protein response. Gene memberships were taken from the mouse MSigDB 2026.1.Mm collections.

Finite signed Wald statistics with nonblank gene symbols were ranked in descending order, without DEG, P-value, FDR, fold-change or expression filtering. Duplicate symbols were resolved by maximum absolute statistic, followed by maximum signed statistic and lexical Ensembl identifier. Each contrast contained 17,441 unique ranked symbols, with no tied statistics.

Enrichment used R 4.5.3 and fgsea 1.36.2 with fgseaMultilevel, weighting exponent 1, gene-set overlaps of 10–500, eps=0 and scoreType=std. The multilevel sample size was 101, with 1,000 preliminary permutations and one process. Seed 20260822 was reset for each contrast using Mersenne–Twister, Inversion and Rejection random-number settings.

Benjamini–Hochberg adjustment covered exactly nine pathways across three contrasts, with significance defined as global FDR < 0.05. Within-contrast FDR values were retained separately. The 27-test bulk family was adjusted separately from each Hallmark–Reactome testing family in GSE168041.

Descriptive expression localization in GSE252285

The GSE252285 analysis included 36,111 cells after per-library quality control. Cells were required to have at least 200 detected genes and satisfy library-specific upper count, feature and mitochondrial thresholds. Upper count and feature thresholds used the smaller of the 99.5th percentile and median plus six scaled median absolute deviations (scaling constant 1.4826; quantile type 8). The mitochondrial threshold used the same rule, bounded between 10% and 30%. Cells were retained irrespective of their doublet classification.

Seurat 5.5.1 normalized counts to 10,000 per cell and applied log transformation [35]. The workflow selected 2,000 variable features using VST, scaled them and calculated 50 principal components. A 20-neighbor graph used components 1–30, and Louvain clustering used resolution 0.6 and seed 42. The original t-SNE used principal components 1–30, perplexity 30 and seed 42; the figures use its stored coordinates. The selected clustering defined 28 clusters.

Broad annotations combined a prior barcode-level annotation with canonical-marker evidence. The majority prior label within each new cluster was calculated among 31,273 cells shared with the earlier annotation. Assigning a named lineage required at least 20 shared cells, a majority-label fraction of at least 0.70 and two supporting canonical markers. Supporting markers required mean log-normalized expression of at least 0.10 and detection in at least 5% of cluster cells. Lineage scores averaged the product of mean expression and detection fraction across each marker panel. Competing-lineage checks used 0.75 of the leading score and a 1.5-fold score advantage over the prior lineage. Clusters with insufficient support or conflicting lineage evidence remained unassigned.

Library assignments were supported by barcode-overlap matching to reconstructed SRA libraries. Suffixes -3, -1 and -2 mapped to GSM7998628 (control), GSM7998629 (hidden hearing loss) and GSM7998630 (noise-induced hearing loss), respectively. The corresponding matched Jaccard indices were 0.9773, 0.9826 and 0.9849. Each assignment was the mutual best match between the deposited suffix group and the reconstructed library. Condition labels for the descriptive summaries were assigned from this external crosswalk.

The twenty-gene panel comprised *Dusp1*, *Fos*, *Nfkbia*, *Arrdc2*, *Apold1*, *Sik1*, *Per1*, *Ccn1*, *Cebpb*, *Fosb*, *Rasd1*, *Gadd45g*, *Depp1*, *Serpine1*, *Errfi1*, *Nr4a1*, *Sgk1*, *Rgs1*, *Egr1* and *Jun*.

Expression detection was defined as a raw count greater than zero. Detection fractions used all cells in the corresponding condition–cell-type group as the denominator. Mean expression used the Seurat RNA data layer and included zero-expression cells.

Groups containing fewer than 20 cells were flagged in the underlying tables. Expression profiles and cell fractions were summarized descriptively for each library. Within each library, broad cell categories were ranked by descending mean normalized expression, with detection fraction used to break ties. Recurrent localization required the same highest-ranked broad cell category in all three libraries and at least 20 cells in each corresponding group.

An annotation sensitivity analysis assigned the 319 cells in cluster 19 to the fibrocyte/mesenchymal category using lineage-marker evidence. The primary annotation retained these cells as unassigned. Gene-expression and UCell summaries were recalculated with the alternative labels, while cell inclusion, expression values, clustering and library assignments remained fixed.

Matched gene set analysis of pathway localization

The six UCell-guided follow-up comparisons used UCell 2.10.1 with maxRank = 1,500 and average ranks for ties [36]. Program memberships used mouse MSigDB 2026.1.Mm gene sets or deterministic unions of those sets. The background comprised 24,428 detected genes. Observed signatures were restricted to detected genes and rescored before comparison with random sets of the same size. The localization statistic was the median score in the target broad cell type minus the median of the other five resolved cell-type medians. Unassigned cells were excluded from this comparison, and condition labels were not used in the statistic.

Random sets matched the detectable signature size and joint mean-expression and detection-rate deciles in a 10×10 grid. Genes were sampled without replacement within the matching strata, excluding members of the tested signature from each candidate pool. Each program used 10,000 iterations with seed 20260823. A sensitivity analysis added GENCODE vM23 gene-span-length tertiles to the matching strata.

The one-sided empirical P value was

$$P = \frac{1 + \#\{T_{\text{null}} \geq T_{\text{observed}}\}}{10{,}000 + 1}.$$

Benjamini–Hochberg adjustment covered the six selected localization comparisons separately for the primary and gene-length sensitivity families [34]. The annotation sensitivity analysis repeated both matching analyses with the alternative cell-type assignment and unchanged scoring, matching, seed and testing families.

Use of artificial intelligence

ChatGPT/Codex (OpenAI) assisted with manuscript preparation, analysis, coding and verification. The author takes responsibility for the final content.

Data Availability

The sequencing datasets are available from the Gene Expression Omnibus under accessions GSE168041, GSE160639 and GSE252285. The manuscript and figure source data are archived in release v1.2.2 at https://github.com/pdx12320/cochlear-noise-transcriptomics. The repository README provides version-pinned links to gene-level and pathway results, sample summaries, expression-localization tables and sensitivity-analysis records. Complete sensitivity bundles remain available in release v1.1.0, and baseline analysis objects in release v1.0.0.

Code Availability

Scripts for GSE168041 preprocessing, annotation, pseudobulk differential expression and enrichment, and for GSE252285 expression localization and matched-null analysis are available in the repository. Current manuscript sources and figure-generation scripts are archived in release v1.2.2 at https://github.com/pdx12320/cochlear-noise-transcriptomics. The repository README links to the raw-count bulk workflows, leave-one-animal-out analyses, annotation-sensitivity analyses and required public inputs. Software versions, input manifests and file checksums accompany the workflows.

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Author contributions

Dengxiang Yang: conceptualization, methodology, software, formal analysis, data curation, visualization, writing—original draft, and writing—review and editing.

Additional Information

Competing interests. The author declares no competing interests.

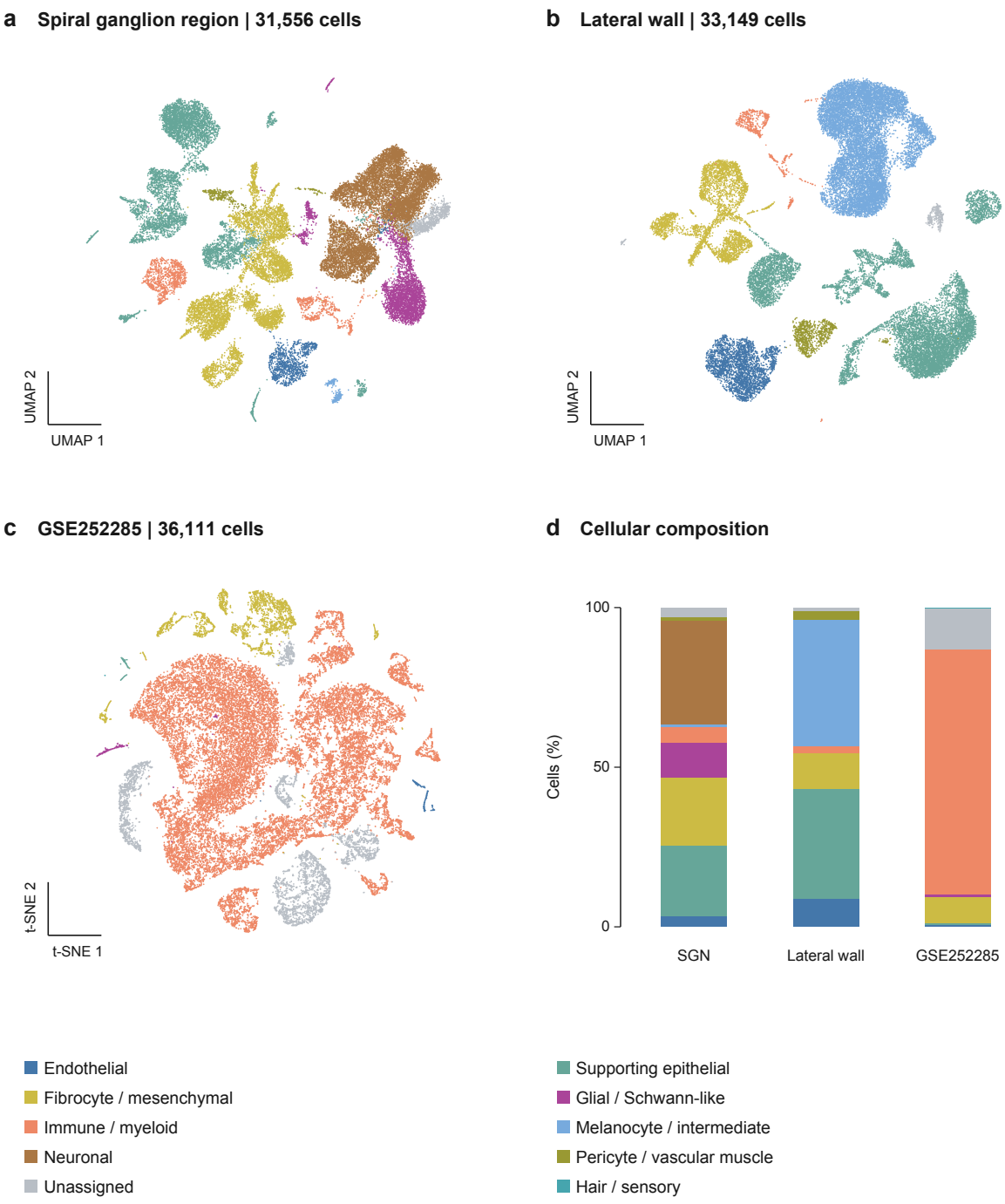


Figure 1: Cochlear cell populations in the two single-cell datasets. (a,b) UMAP projections of 31,556 spiral-ganglion-region cells and 33,149 lateral-wall cells from GSE168041, coloured by broad cell annotation. (c) The t-SNE projection of 36,111 GSE252285 cells using the selected resolution-0.6 clustering and primary broad annotations. Each point represents one cell. (d) Cell-type composition within the two GSE168041 compartments and GSE252285. Colours identify the same broad categories across panels; fractions use all retained cells as denominators. GSE168041 includes four naive and four noise-exposed samples per compartment. GSE252285 includes Control, HHL and NIHL libraries. SGN, spiral ganglion region; HHL, hidden hearing loss; NIHL, noise-induced hearing loss.

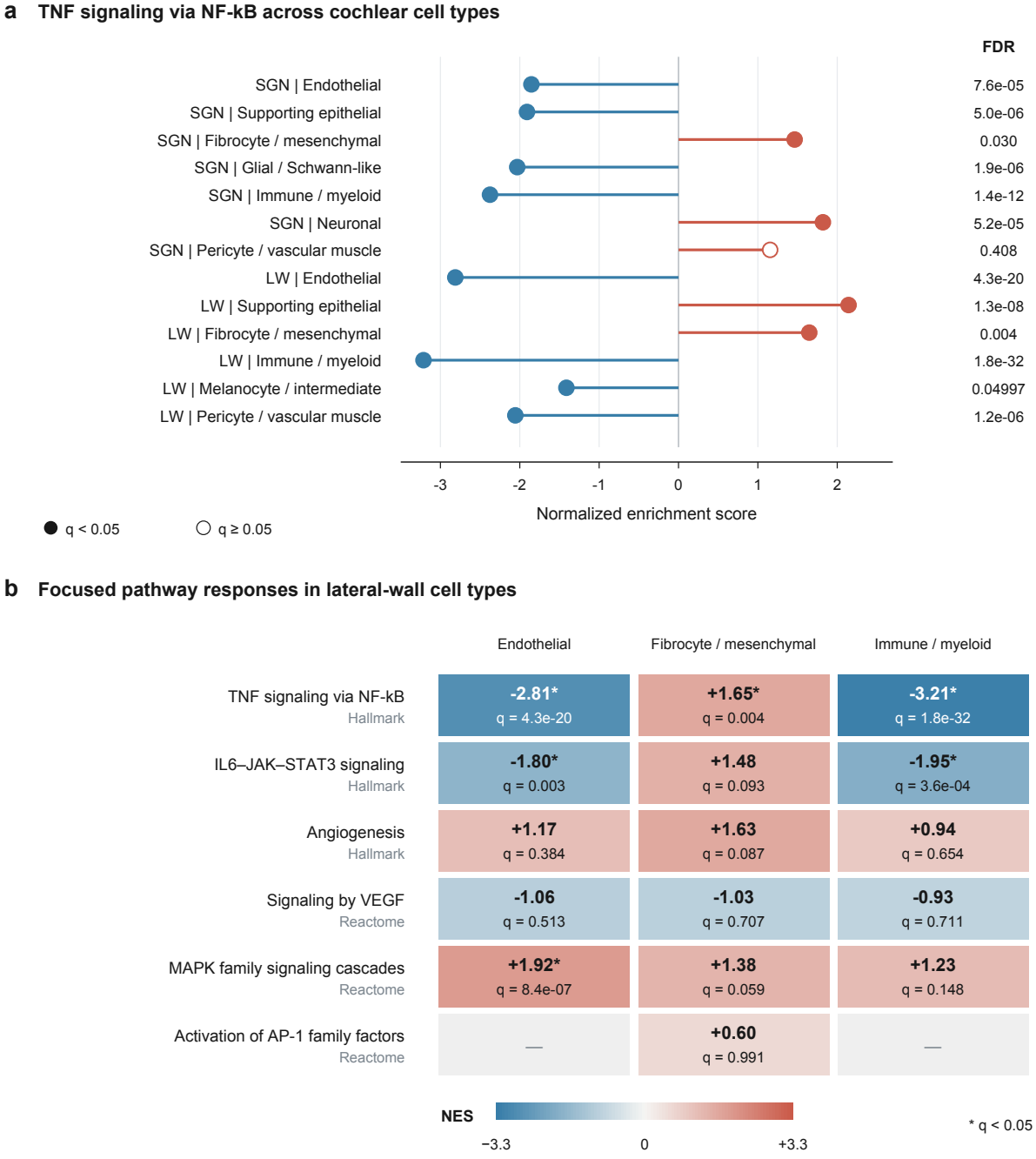


Figure 2: Cell-type-specific pathway responses to noise exposure. (a) Normalized enrichment scores (NES) for the Hallmark TNF signaling via NF-κB gene set across all thirteen eligible tissue–cell-type combinations in GSE168041. Filled points denote FDR < 0.05. Blue and red indicate enrichment toward naive and noise-exposed samples, respectively. The right-hand column gives FDR values. (b) NES and FDR values for six focused Hallmark and Reactome sets in lateral-wall endothelial, fibrocyte/mesenchymal and immune/myeloid cells. Asterisks identify FDR < 0.05; grey cells with a dash mark gene sets outside the eligible size range. Hallmark angiogenesis, Reactome VEGF signaling, Reactome MAPK family signaling cascades and Reactome AP-1 activation are reported separately. Each sample-level pseudobulk contrast used four naive and four noise-exposed samples. LW, lateral wall; SGN, spiral ganglion region.

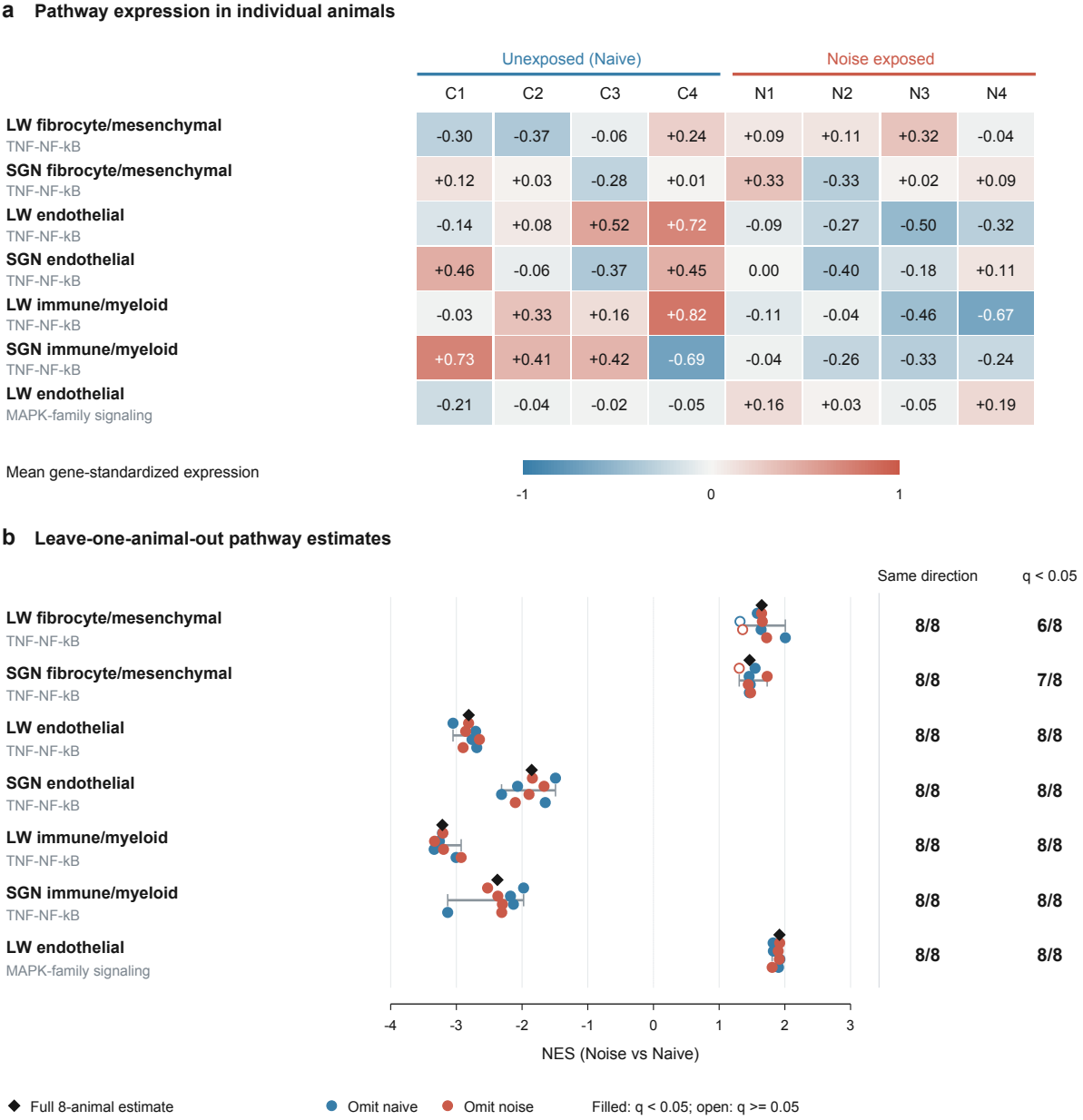


Figure 3: Individual-sample profiles and sensitivity of the core pathway responses. (a) Standardized pathway-expression profiles for the six TNF–NF-κB tissue–cell-type combinations and lateral-wall endothelial MAPK signaling. Normalized pseudobulk counts were log1p-transformed, standardized within each gene across the eight samples, and averaged over the fixed mapped pathway members. C1–C4 and N1–N4 denote naive and noise-exposed samples, respectively; codes are indexed separately within each tissue. (b) Full-sample and leave-one-sample-out NES estimates for the same seven pathway–model combinations. Each model used all eight samples for its main estimate and seven samples for each of eight omissions. Black diamonds show the full-sample estimates. Blue and red circles denote omission of a naive or noise-exposed sample, respectively; filled circles mark FDR < 0.05. Horizontal lines show the minimum-to-maximum range of the eight omission estimates. The columns report direction agreement with the full model and the number of omission fits with FDR < 0.05. All originally eligible Hallmark and Reactome sets were included in each model's multiple-testing family. LW, lateral wall; SGN, spiral ganglion region; NES, normalized enrichment score; FDR, false discovery rate.

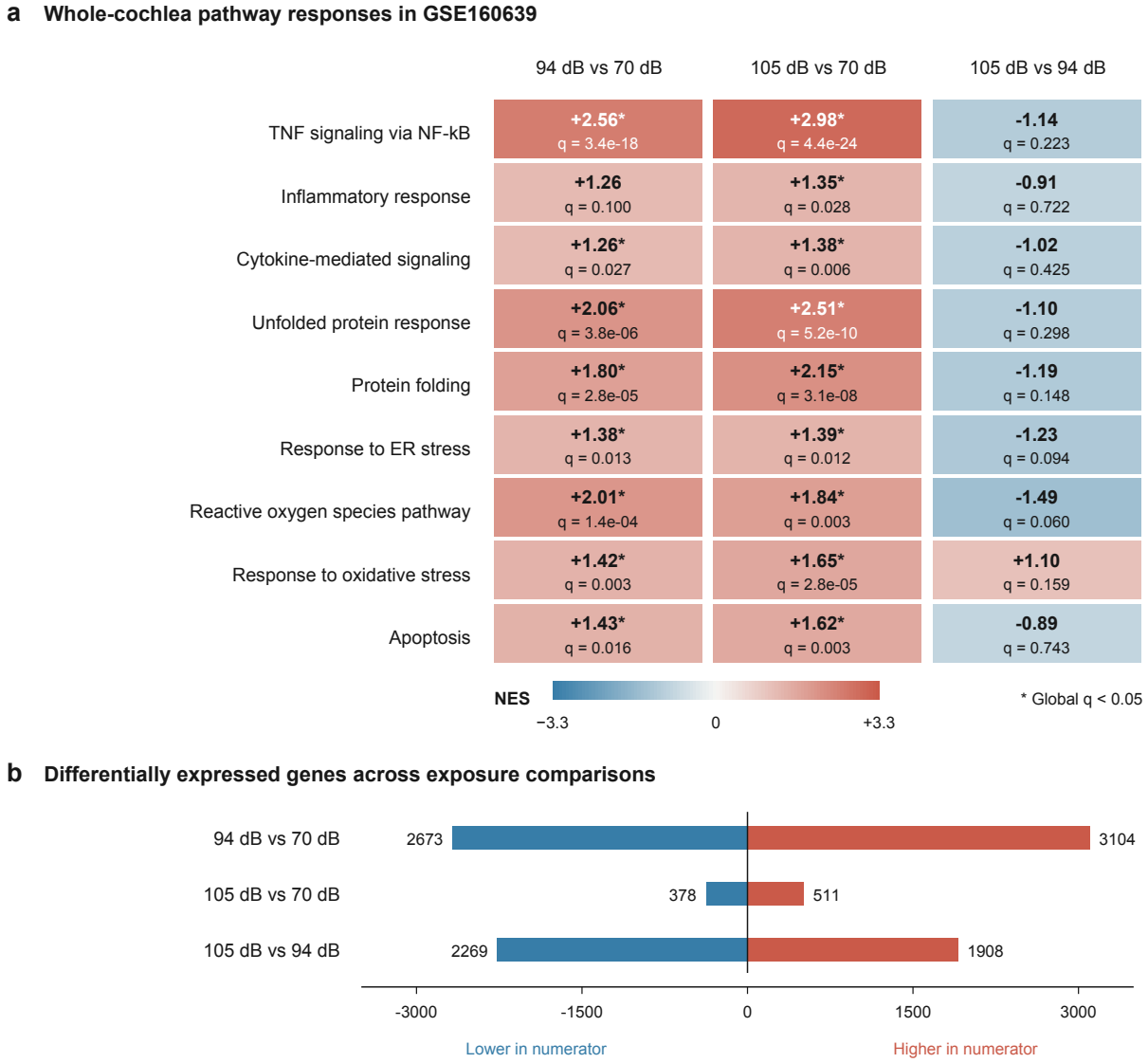


Figure 4: Whole-cochlea transcriptional responses across noise exposures. (a) All nine targeted gene sets across the three GSE160639 exposure comparisons, estimated from raw-count DESeq2 contrasts. Each tile reports the normalized enrichment score (NES) and the Benjamini–Hochberg FDR adjusted jointly across all 27 tests. Positive NES indicates enrichment toward the first exposure in the stated comparison. Asterisks identify global FDR < 0.05. Hallmark sets comprise TNF signaling via NF-κB, inflammatory response, unfolded protein response, reactive oxygen species pathway and apoptosis; the remaining sets are Gene Ontology Biological Process terms. (b) Gene counts at FDR < 0.05 from DESeq2 Wald tests, with blue and red bars denoting lower and higher expression in the numerator exposure. No additional fold-change threshold was applied. Gene-level and pathway-level FDR values refer to their respective multiple-testing families.

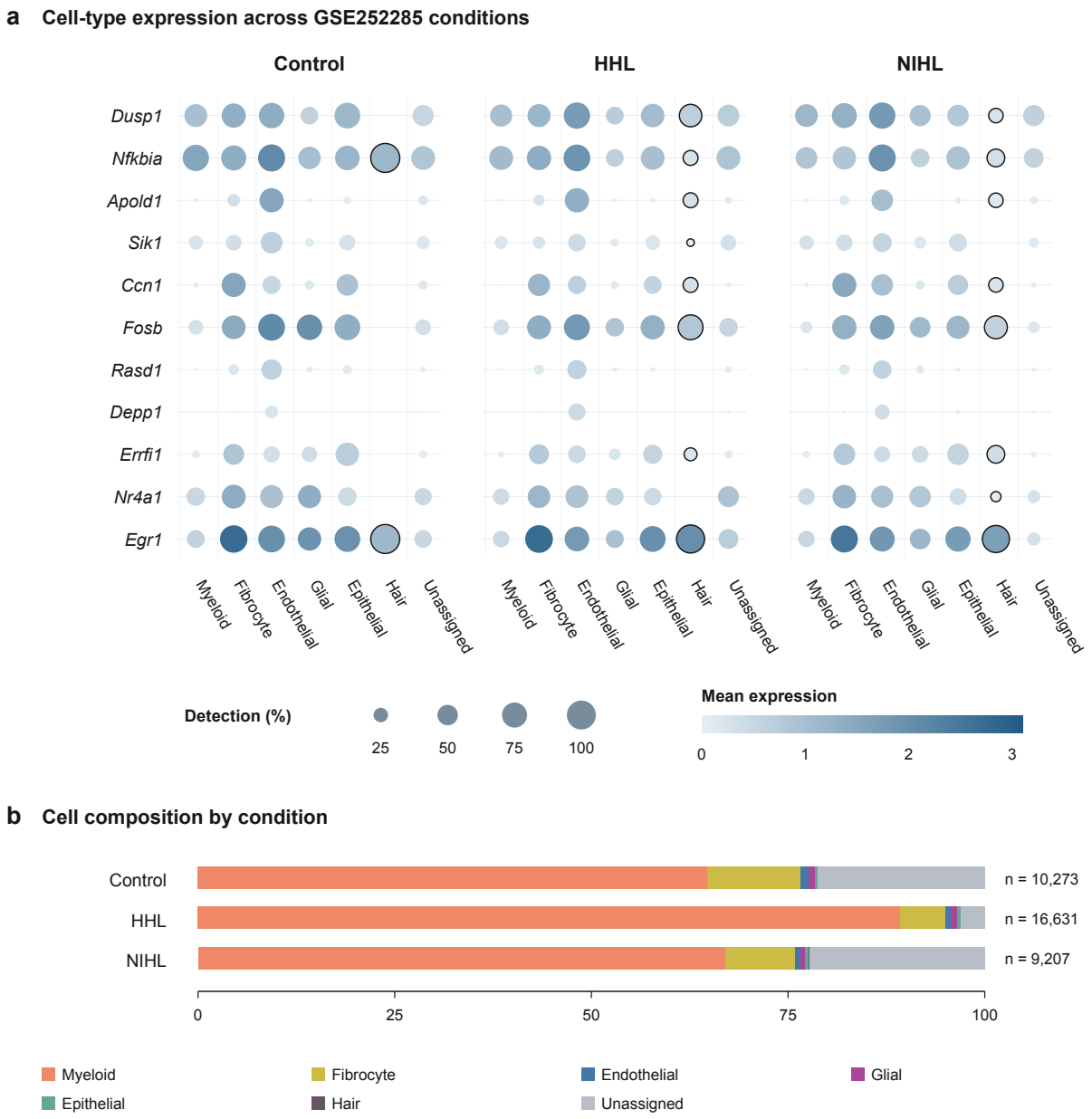


Figure 5: Condition-resolved expression of genes with stable cell-type localization. (a) Eleven genes whose highest-ranked broad cell type by mean normalized expression (with detection fraction used to break ties) was retained across Control, HHL and NIHL conditions in GSE252285, with at least 20 cells in each top-ranked group. Dot area represents the percentage of cells with RNA counts greater than zero. Colour represents mean normalized expression across all cells in each group, including zeros, on a shared scale. Outlined dots identify groups containing fewer than 20 cells. (b) Broad cell-type composition and cell counts in the three condition libraries. The three libraries contain 36,111 cells, summarized using their existing clusters and annotations. Fibrocyte, epithelial and hair labels denote fibrocyte/mesenchymal, supporting/epithelial and hair/sensory populations, respectively. HHL, hidden hearing loss; NIHL, noise-induced hearing loss.



Figure 6: Cluster 19 expresses mesenchymal markers and contributes to *Apold1* expression. (a) The fixed GSE252285 t-SNE projection highlights primary endothelial and fibrocyte/mesenchymal populations and the 319 cells in cluster 19. (b) Canonical-marker expression in endothelial cells, primary fibrocyte/mesenchymal cells and cluster 19. Dot area denotes the detection fraction, and colour denotes mean log-normalized expression on a shared scale. (c,d) Mean log-normalized *Apold1* expression and detection fractions in pooled cells and each library. Primary fibrocyte/mesenchymal summaries exclude cluster 19; the reassigned summaries include its 319 cells. Cluster 19 is also shown separately to retain its distinct expression profile. Endothelial summaries use the same 275 cells under both annotation schemes. Each bar summarizes cells within its stated group. HHL, hidden hearing loss; NIHL, noise-induced hearing loss.